CollaRAG: Collaborative Multi-Agent Retrieval via Division of Labor

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Abstract

Existing Retrieval-Augmented Generation (RAG) systems apply a uniform retrieval pipeline to all queries, regardless of whether a question demands a single fact lookup or multi-hop reasoning across documents. This monolithic design forces a compromise: either invest heavy computation on every query, or under-serve complex ones. We introduce COLLARAG, a multi-agent RAG framework built on the principle of *division of labor*. COLLARAG makes three contributions. **First**, we propose *Intent-Aware Query Triage*, a two-level classifier that routes each query to the best-suited specialist agent, eliminating the one-pipeline-fits-all bottleneck. **Second**, we design a *Plan-Execute-Reflect* workflow for the multi-hop agent, which decomposes complex queries into structured retrieval sub-plans, executes each through a bounded tool-use loop, and self-reflects on evidence gaps before synthesis. **Third**, we introduce *Dual-Layer Experiential Memory*, pairing a cross-agent shared memory (caching, routing statistics, entity aliases) with per-agent private memory that accumulates successful retrieval patterns over time. Five specialized agents share a pool of 22 retrieval, reasoning, and memory tools, with each agent's access governed by a static tool-permission matrix that enforces capability specialization. On six QA benchmarks spanning single-hop (NQ, PopQA), multi-hop (HotpotQA, 2WikiMultiHopQA, MuSiQue), and long-form summarization (ASQA) tasks, COLLARAG sets new state-of-the-art results under both Qwen3-8B and Qwen3-32B backbones, outperforming the previous best method HippoRAG2 [Gutierrez et al., 2025] by up to 5.7 F1 on 2WikiMultiHopQA (Qwen3-8B) and 5.1 F1 on HotpotQA (Qwen3-32B), with the largest gain of 7.2 F1 on ASQA (Qwen3-8B), where the dedicated Map-Reduce agent has no baseline counterpart. Ablation studies confirm that intent-aware routing is the single most impactful component, contributing 5.9–12.2 EM points when removed.

1 Introduction

Large language models (LLMs) have achieved remarkable performance across a wide range of language understanding and generation tasks [Wei et al., 2022]. Despite their success, LLMs suffer from three well-documented limitations that hinder reliable deployment in knowledge-intensive applications. First, *knowledge staleness*: the parametric knowledge of an LLM is frozen at its training cutoff, rendering it unable to reflect recent events or updated domain facts. Second, *hallucination*: LLMs can generate fluent but factually incorrect content, confidently presenting fabricated information indistinguishable from genuine knowledge [Gao et al., 2024]. Third, *provenance opacity*: generated answers lack source attribution, making it impossible for users to verify claims or trace them back to authoritative evidence. These shortcomings are especially problematic in enterprise question answering, legal and medical consultation, and scientific literature review, where factual accuracy and traceability are non-negotiable requirements.

Retrieval-Augmented Generation (RAG) [Lewis et al., 2020] has emerged as a leading paradigm for addressing these limitations. By retrieving relevant passages from an external knowledge base and prepending them to the LLM prompt at inference time, RAG grounds generation in verifiable evidence without retraining the model. This architecture offers three practical advantages: (1) knowledge can be updated by modifying the external corpus rather than retraining billion-parameter models; (2) generated answers can be traced back to specific source passages, enabling auditability; (3) the same LLM can serve multiple domains by swapping the underlying knowledge base. Backed by mature infrastructure—vector

databases such as FAISS [Johnson et al., 2019] and Milvus, embedding models [Chen et al., 2024], and orchestration frameworks like LangChain and LlamaIndex—RAG has become one of the most widely adopted paradigms for deploying LLMs in knowledge-intensive production systems.

The standard RAG pipeline follows a sequential four-stage architecture: *chunking* (splitting documents into passages), *indexing* (encoding chunks into dense vectors), *retrieval* (finding top-k relevant chunks via similarity search), and *generation* (feeding retrieved chunks to the LLM for answer synthesis). Existing RAG methods have advanced the retrieval substrate along three directions: *text-based* methods enhance passage retrieval through hierarchical indexing (e.g., RAPTOR [Sarthi et al., 2024]) and hybrid dense-sparse fusion; *graph-based* methods such as TOG [Sun et al., 2024] and G-Retriever [He et al., 2024] exploit entity-relation structures for multi-hop reasoning; and *hybrid* methods—GraphRAG [Edge et al., 2024], LightRAG [Guo et al., 2024], HippoRAG [Gutierrez et al., 2024]—combine textual evidence with graph traversal, achieving current state-of-the-art results on multi-hop QA benchmarks.

Yet across all three families, a fundamental design assumption remains unchallenged: *every query passes through the same retrieval pipeline*. Whether a user asks a trivial factual question or a complex comparative analysis requiring evidence chains across multiple documents, the system invokes identical retrieval logic, tool budgets, and reasoning strategies. We term this pervasive pattern *retrieval monolithism*.

Retrieval monolithism creates three concrete problems:

- **Simple query over-processing.** Single-hop factual questions that need only one passage lookup are funneled through sophisticated multi-step pipelines (e.g., GraphRAG's community summarization or HippoRAG's PageRank traversal), incurring unnecessary latency and token cost without quality improvement. In mixed query workloads that include factoid, multi-hop, and summarization requests, a substantial fraction of queries are single-hop, making uniform heavy pipelines inefficient.
- **Complex query under-reasoning.** Multi-hop questions require chaining evidence across multiple documents, often involving entity comparison, temporal reasoning, or causal inference. Existing methods apply the same retrieval budget regardless of query complexity. Even graph-based systems typically perform retrieval in a single forward pass without iterative evidence verification: when the initial retrieval misses a critical linking entity, most systems lack a mechanism to detect and repair the gap.
- **Summarization query under-aggregation.** Queries requesting overviews or syntheses require global information aggregation across many passages, not point retrieval of specific facts. Existing multi-hop QA-oriented RAG systems rarely treat summarization as a first-class routed intent with a dedicated aggregation pipeline. As a result, summarization requests are often handled by the same top-k retrieval mechanism used for factoid QA, which may cover only a small portion of the relevant corpus and produce incomplete summaries.

These limitations are structural consequences of the monolithic design, not bugs in individual systems. Addressing them requires rethinking the retrieval architecture itself.

We draw inspiration from the *division of labor* principle. In organizational design, a hospital triages patients by symptom severity, routing a headache to general medicine and a fracture to orthopedics. Each department has specialized equipment and protocols optimized for its patient population. No hospital sends every patient through the same diagnostic pipeline. Similarly, an intelligent retrieval system should route queries to specialized agents based on query characteristics, equipping each agent with the tools and reasoning strategies best suited to its query type.

We introduce COLLARAG (**Colla**borative Multi-Agent **RAG**), a framework that replaces the monolithic retrieval pipeline with intent-conditioned specialist execution. COLLARAG routes queries into five handling modes—greeting short-circuit, SingleHop, MultiHop, Summarization, and Other—supported by a Preprocessor agent and four task-specialist agents. An *intent-aware query triage* module classifies incoming queries and dispatches each to the matching handler. Each task agent draws from a shared pool of 22 retrieval, reasoning, and memory tools, with access governed by a static *tool-permission matrix* that

enforces capability specialization. The MultiHop agent employs a novel *Plan-Execute-Reflect* workflow. A *dual-layer memory system* enables cross-agent coordination and per-agent progressive learning.

Specifically, COLLARAG makes three technical contributions:

1. **Intent-Aware Query Triage** (§3.1). We propose a two-level intent classifier—a fast regex-based greeting filter cascading into LLM semantic classification with keyword-based rule fallback—that categorizes queries into five handling modes (`greeting`, `single_hop`, `multi_hop`, `summarization`, `other`) and dispatches each to the matching handler via one-shot deterministic routing. Unlike prior adaptive retrieval methods that mainly switch among retrieval depths or strategy templates, our triage operates at the agent level: different intents are handled by specialists with distinct tool permissions, reasoning protocols, synthesis procedures, and private memories.
2. **Plan-Execute-Reflect Workflow** (§3.2). For complex multi-hop queries, we design a three-phase workflow that separates *what to find* (Plan), *how to find it* (Execute), and *whether it is sufficient* (Reflect). The Plan phase generates a structured JSON step list; the Execute phase runs each step through a bounded mini-ReAct loop [Yao et al., 2023]; the Reflect phase assesses evidence quality and triggers targeted supplementary retrieval before final synthesis. This design addresses the two key weaknesses of flat ReAct loops: absence of explicit planning and lack of systematic gap detection.
3. **Dual-Layer Experiential Memory** (§3.3). A *shared memory* layer provides cross-agent coordination through QA caching, routing statistics, and entity alias resolution. An isolated *agent private memory* stores per-agent retrieval experience—successful tool invocation sequences, quality scores, and failure lessons—with deduplication and cold-eviction policies. This dual-layer architecture enables progressive learning while maintaining strict agent isolation for clean ablation.

We evaluate COLLARAG on six QA benchmarks that collectively span three intent types: single-hop factoid QA (NQ [Kwiatkowski et al., 2019], PopQA [Mallen et al., 2023]), multi-hop reasoning (HotpotQA [Yang et al., 2018], 2WikiMultiHopQA [Ho et al., 2020], MuSiQue [Trivedi et al., 2022]), and long-form summarization (ASQA [Stelmakh et al., 2022]). This mixed-intent evaluation directly validates the core hypothesis: a system designed for division of labor should excel when queries demand different retrieval strategies. Using both Qwen3-8B and Qwen3-32B as backbone LLMs, COLLARAG achieves state-of-the-art results across all benchmarks and backbone scales, outperforming the previous best method HippoRAG2 [Gutierrez et al., 2025] by up to 5.7 F1 on 2WikiMultiHopQA under Qwen3-8B and 5.1 F1 on HotpotQA under Qwen3-32B. Comprehensive ablation studies confirm that intent-aware routing is the single most impactful component (removing it causes -11.7 EM on 2WikiMultiHopQA), validating the division-of-labor hypothesis.

2 Related Work

2.1 RAG Foundations and Retrieval Enhancement

RAG [Lewis et al., 2020] established the retrieve-then-generate paradigm, pairing parametric LLMs with non-parametric passage retrieval to ground generation in external evidence. Many vanilla RAG implementations rely on DPR-style bi-encoder retrieval [Karpukhin et al., 2020] for open-domain QA, and Fusion-in-Decoder [Izacard and Grave, 2021] scaled this approach to processing many retrieved passages in parallel. Subsequent work has refined the retrieval stage through multiple dimensions: hybrid retrieval combining dense and sparse (BM25 [Robertson and Zaragoza, 2009]) signals via reciprocal rank fusion [Cormack et al., 2009]; cross-encoder reranking [Nogueira and Cho, 2020] for more precise relevance estimation; and query rewriting [Ma et al., 2023] for bridging the vocabulary gap between queries and documents. RAPTOR [Sarthi et al., 2024] organizes retrieval via recursive abstractive trees, enabling retrieval at multiple abstraction levels. These methods enhance retrieval quality but apply a

uniform pipeline to all queries regardless of complexity—the monolithic limitation that CollaRAG addresses.

2.2 Graph-based and Hybrid RAG

Knowledge graphs provide structured pathways for multi-hop reasoning. GraphRAG [Edge et al., 2024] performs community detection on entity graphs using the Louvain algorithm [Blondel et al., 2008] and generates hierarchical summaries for query-focused retrieval. LightRAG [Guo et al., 2024] introduces dual-level (key-entity and high-level topic) graph retrieval for improved efficiency. TOG [Sun et al., 2024] uses LLMs to iteratively traverse knowledge graph triples, while G-Retriever [He et al., 2024] integrates graph neural networks with LLMs for subgraph retrieval. HippoRAG [Gutierrez et al., 2024] draws on hippocampal indexing theory, implementing personalized PageRank over a knowledge graph to mimic human associative memory, and HippoRAG2 [Gutierrez et al., 2025] extends this with deeper memory integration and passage-level retrieval. These systems optimize *what* is retrieved by enriching the knowledge substrate, but do not adapt *how* retrieval is orchestrated based on query type—an orthogonal dimension that CollaRAG addresses through intent-aware routing. CollaRAG is complementary to these retrieval substrates: graph traversal, dense retrieval, and hybrid retrieval can all serve as tools inside specialist agents.

2.3 Adaptive RAG and Query Routing

Recent work has begun to challenge the one-size-fits-all assumption. Self-RAG [Asai et al., 2024] trains an LLM to decide *when* to retrieve and to self-critique generated outputs using reflection tokens, but applies the same retrieval mechanism once triggered. Adaptive-RAG [Jeong et al., 2024] trains a query-complexity classifier (on T5-Large) to route among three strategies: no retrieval, single-step retrieval, and multi-step retrieval. CRAG [Yan et al., 2024] evaluates retrieval quality and triggers corrective web searches when confidence is low. These approaches adapt retrieval decisions at the strategy level—whether to retrieve, how many retrieval steps to take, or whether to correct retrieval results—but route to pre-defined strategies within a single pipeline. CollaRAG performs routing at the agent level: the selected handler differs not only in retrieval depth but also in tool permissions, reasoning protocol (single-round vs. Plan-Execute-Reflect vs. Map-Reduce), memory scope, and synthesis procedure, realizing a true division of labor rather than a strategy-selection switch.

2.4 Multi-hop Reasoning and Query Decomposition

Multi-hop QA requires chaining evidence across multiple passages. Chain-of-thought prompting [Wei et al., 2022] enables LLMs to articulate intermediate reasoning steps. IRCoT [Trivedi et al., 2023] interleaves retrieval with chain-of-thought steps, using each reasoning step to guide the next retrieval query, achieving strong results on HotpotQA, 2Wiki, and MuSiQue. Self-Ask [Press et al., 2023] and Least-to-Most prompting [Zhou et al., 2023] decompose complex questions into simpler sub-questions that can be answered sequentially. Plan-and-Solve [Wang et al., 2023] generates explicit plans before execution. These methods improve multi-hop reasoning within a single pipeline. CollaRAG's MultiHop agent builds on these ideas but introduces a key structural innovation: separating the plan, execute, and reflect phases into distinct stages, each with bounded iteration costs and an explicit quality gate before synthesis.

2.5 Agentic RAG and Multi-Agent LLM Systems

The rise of LLM-based agents [Xi et al., 2023] has introduced tool use into RAG. ReAct [Yao et al., 2023] interleaves reasoning and action (tool invocation) in a single loop. Toolformer [Schick et al., 2023] teaches LLMs to decide when and how to call external tools. Multi-agent frameworks such as MetaGPT [Hong

et al., 2024] and AutoGen [Wu et al., 2023] enable multiple agents to collaborate through conversational protocols. However, these multi-agent systems assume agents need to *negotiate*—exchanging messages, debating solutions, or voting on outcomes. COLLARAG adopts a fundamentally different *dispatch* model: a classifier makes a one-shot routing decision, and the assigned specialist agent executes independently without inter-agent dialogue. This design prioritizes deterministic reproducibility and clean ablation boundaries over emergent collaboration, while still achieving the benefits of specialization through the tool-permission matrix.

2.6 Memory-Augmented RAG

Standard RAG systems are stateless: each query is processed independently with no memory of past interactions. HippoRAG [Gutierrez et al., 2024] introduced biologically-inspired long-term memory into RAG through knowledge graph-based indexing. MemoRAG [Qian et al., 2025] builds a global memory model that recalls query-specific clues to guide retrieval over long contexts. ExpeL [Zhao et al., 2024] demonstrates that LLM agents can learn from accumulated experience, extracting insights from past task trajectories to improve future performance. Reflexion [Shinn et al., 2023] stores verbal self-reflections as episodic memory to prevent repeating mistakes. COLLARAG contributes a *dual-layer* memory architecture that separates cross-agent shared memory (QA caching, routing statistics, entity aliases) from per-agent private experiential memory (tool sequences, quality scores, lessons learned). This separation enables both collaborative knowledge sharing and isolated experiential learning, with clean ablation boundaries between the two layers.

2.7 Retrieval-Augmented Summarization

While most RAG systems focus on factoid QA, real-world queries often demand summarization or synthesis across many passages. The Map-Reduce paradigm for distributed text processing [Dean and Ghemawat, 2008] has been adapted to LLM-based summarization: each passage is independently summarized (Map), then all summaries are aggregated into a coherent overview (Reduce). GraphRAG [Edge et al., 2024] uses community-level summaries for global queries but does not offer a dedicated summarization agent. Existing RAG surveys [Gao et al., 2024, Xu et al., 2024] identify summarization as an underserved query type in current systems. COLLARAG addresses this gap with a dedicated Summarization agent that implements a full Map-Reduce pipeline: broad retrieval ($k = 50$), per-chunk LLM summarization, and final aggregation. This makes summarization a first-class routed intent within the same multi-agent RAG framework, rather than an external summarization chain or a generic retrieval mode.

3 Method

COLLARAG is built on a single thesis: a retrieval system should not push every query through the same pipeline, because queries differ in the amount of evidence, the shape of reasoning, and the kind of aggregation they require. We realize this thesis as an orchestrator that, for each query, decides *which* specialist should handle it, equips that specialist with a bounded set of tools, and lets it follow a protocol matched to the query type. Figure 1 sketches the end-to-end flow. Concretely, a query passes through six stages: (i) intent classification, (ii) a greeting short-circuit, (iii) a QA-cache lookup, (iv) query preprocessing, (v) dispatch to one of five specialist agents, and (vi) post-processing that writes back to memory. The three contributions of this paper sit at the decision points of this flow: how a query is routed (§3.1), how the hardest queries are reasoned about (§3.2), and how experience accumulates across queries (§3.3).

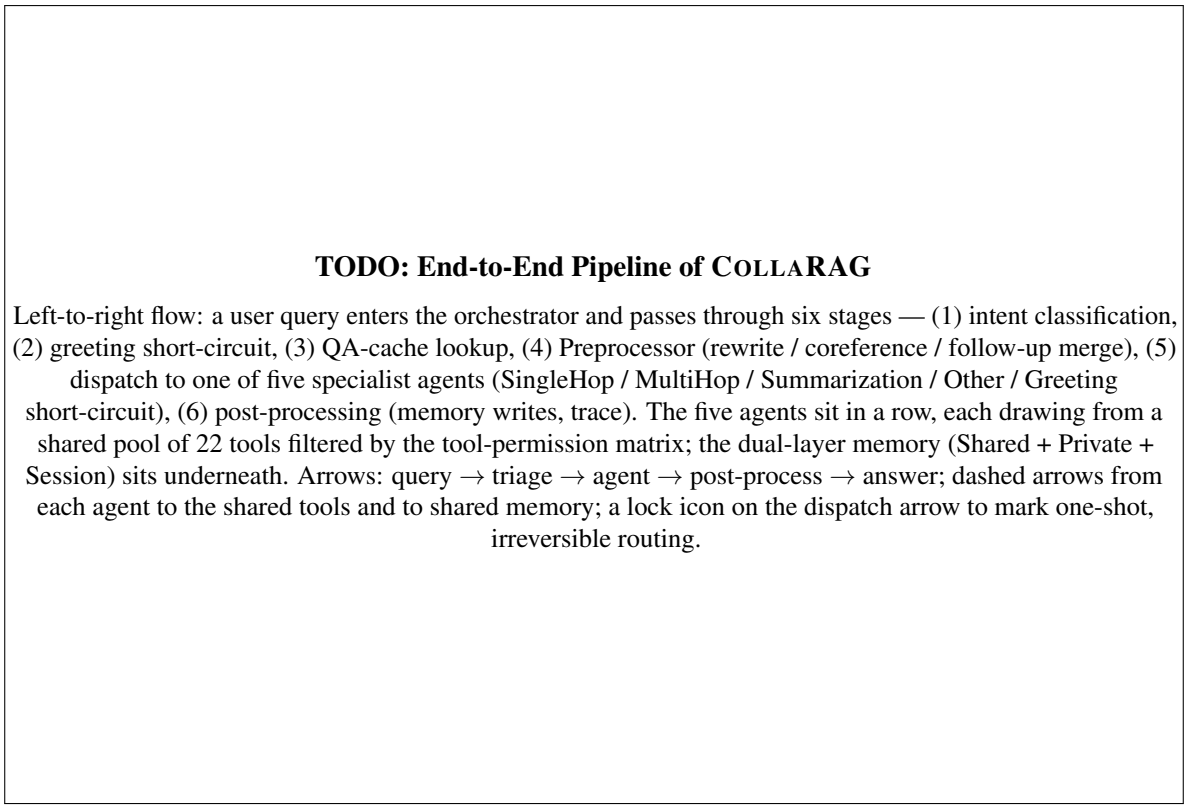


Figure 1: End-to-end pipeline of CollaRAG. The orchestrator routes each preprocessed query to one specialist agent; agents share a tool pool and a dual-layer memory. (Figure to be drawn.)

3.1 Intent-Aware Query Triage

The triage module answers one question per query: which specialist should run? It is a two-level cascade rather than a single classifier, because greetings are trivially detectable and should never pay the cost of a semantic LLM call. Level 1 is a regex filter for greeting patterns; a hit returns a direct LLM response and skips the rest of the pipeline. Level 2 sends the non-greeting query to the backbone LLM, which returns one of four labels—`single_hop`, `multi_hop`, `summarization`, or `other`—and a keyword rule fallback covers parse failures (summarization and comparative keywords, else `single_hop`). Formally, the routing decision is

$$a^*(q) = \text{dispatch}\big(\underbrace{L_2(q)}_{\text{LLM label}} \circ \underbrace{L_1(q)}_{\text{regex}}\big), \quad a^* \in \{\text{SH}, \text{MH}, \text{SM}, \text{OT}\} \cup \{\text{greet}\}, \tag{1}$$

where L_1 short-circuits greetings and L_2 labels the rest. Routing is one-shot and irreversible: once a^* is chosen, the query stays with that agent. We impose this on purpose, so that each specialist's marginal contribution can be measured cleanly (§5) and so that a single misclassification cannot cascade through re-routing.

Five specialists. Each label selects an agent whose protocol fits the query's demand profile. SingleHop runs one round of hybrid (dense + BM25) retrieval, fuses the two rank lists by reciprocal rank fusion,

$$s(d) = \sum_{i \in \{\text{vec}, \text{bm25}\}} \frac{1}{\text{rank}_i(d) + k}, \tag{2}$$

optionally reranks, and generates after a one-shot quality check—the cheapest path, sufficient for a single-fact lookup. MultiHop runs the Plan-Execute-Reflect workflow of §3.2. Summarization runs a

TODO: Two-Level Query Triage

A decision tree. Root: query q. Level 1: regex greeting filter $\rightarrow$ on hit, short-circuit to a direct LLM answer. Level 2 (non-greeting): an LLM semantic classifier with a dashed fallback edge for parse failures, branching into four leaves: `single_hop`→SingleHop, `multi_hop`→MultiHop, `summarization`→Summarization, `other`→Other. Annotate the one-shot, irreversible dispatch with a lock icon on the leaf edges.

Figure 2: Two-level query triage. L_1 short-circuits greetings; L_2 labels the rest into four specialist routes. (Figure to be drawn.)

Tool Category	**PP**	**SH**	**MH**	**SM**	**OT**
Retrieval (6)	0	4	6	4	1
Entity (3)	0	2	3	1	1
Reasoning (3)	0	2	3	1	0
Preprocessing (3)	3	0	0	0	0
Memory (3 read)	2	2	3	2	2
Web (2)	0	0	0	0	2
IO (2)	0	2	2	2	2
Total	5	12	17	10	8

Table 1: Tool-permission matrix. PP=Preprocessor, SH=SingleHop, MH=MultiHop, SM=Summarization, OT=Other. The MultiHop agent has the richest toolkit (17/22), reflecting its need for graph traversal, sub-query decomposition, and evidence assessment.

Map-Reduce pipeline that retrieves a broad pool ($k = 50$), summarizes each passage locally, and reduces the summaries into one overview, a shape of aggregation no factoid-retrieval pipeline provides. Other falls back to web search with optional local retrieval for out-of-domain queries. A Preprocessor agent rewrites the query (expansion, coreference resolution, follow-up merging) before triage; it holds no retrieval tools and only cleans the query text.

Tool-permission matrix. All five agents share one pool of 22 tools, but each agent sees only a curated subset (Table 1). The matrix is the infrastructure-level enforcement of specialization: MultiHop alone can call graph path search and sub-query decomposition, while SingleHop works with a focused 12-tool set. This prevents a nominally specialized agent from quietly behaving like a generalist.

3.2 Plan-Execute-Reflect Workflow

Multi-hop queries are the hardest case: the answer depends on a chain of facts spread across passages, and a single retrieval pass usually returns one link without the others. A flat ReAct loop [Yao et al., 2023] interleaves thought and action greedily, which (i) never commits to an explicit plan and (ii) has no

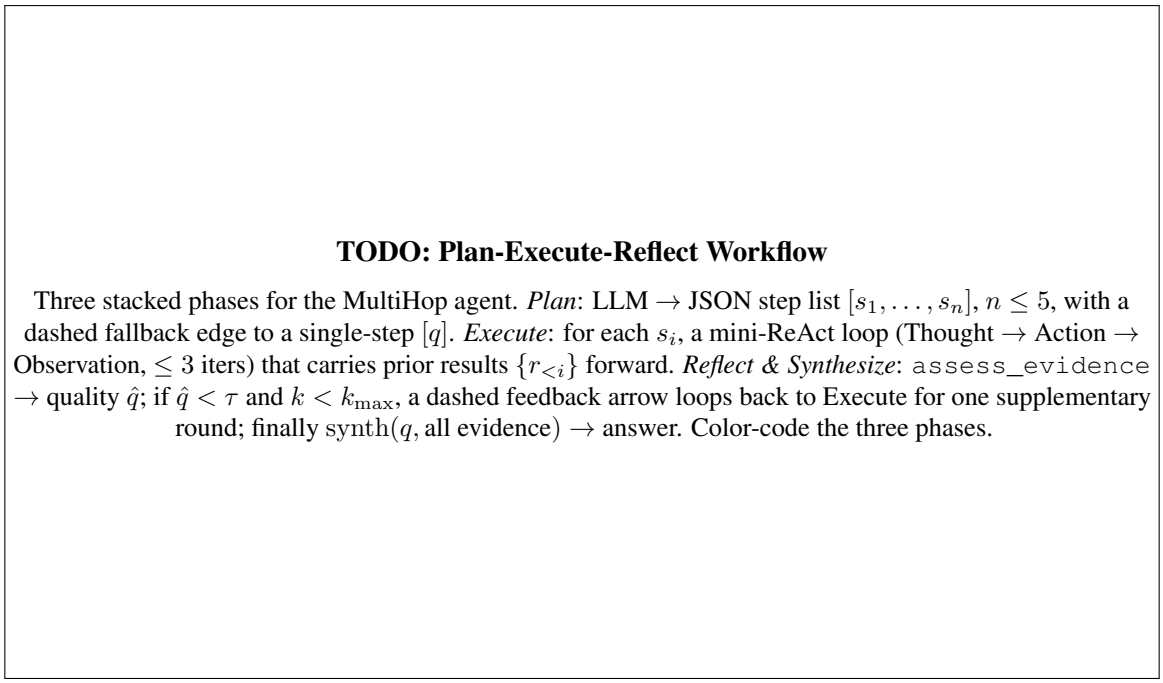


Figure 3: The Plan-Execute-Reflect workflow of the MultiHop agent. Plan decides *what* to find, Execute *how*, Reflect *whether it is enough*. (Figure to be drawn.)

principled way to notice that evidence is incomplete. The Plan-Execute-Reflect (PER) workflow separates these two concerns from execution itself, into three phases.

Plan. Given q, the LLM produces a retrieval plan as a bounded JSON list of sub-tasks,

$$P = \big[\{s_1, T_1\}, \ldots, \{s_n, T_n\}\big], \qquad n \le n_{\max}{=}5, \tag{3}$$

where s_i is a sub-query and T_i a tool hint. If the output is unparseable, the plan degrades to $[\{q, \emptyset\}]$, so a malformed plan never blocks the pipeline—the agent still answers, just without decomposition.

Execute. Each s_i runs as a mini-ReAct loop that may call tools and that sees the results of all earlier steps, so evidence flows forward across steps rather than being rediscovered:

$$r_i = \mathrm{MiniReAct}\big(s_i,\, q,\, \{r_1, \ldots, r_{i-1}\},\, \mathcal{T}_{\mathrm{MH}}\big), \qquad \text{at most } j_{\max}{=}3 \text{ iterations.} \tag{4}$$

Bounding $j_{\max}$ keeps the cost of any single step predictable, which an unbounded ReAct loop does not.

Reflect and synthesize. After all steps finish, an evidence assessor scores the gathered evidence $\hat{q} = \texttt{assess_evidence}(q) \in [0, 1]$. If $\hat{q} < \tau$ (default $\tau{=}0.6$) and the reflection budget $k < k_{\max}{=}2$ remains, a reflection prompt pinpoints the missing piece and triggers one supplementary mini-ReAct round; otherwise the loop stops. The final answer synthesizes everything:

$$\text{answer} = \mathrm{LLM}_{\mathrm{synth}}\big(q,\, \{r_1, \ldots, r_n\} \cup \{\mathrm{refl}_1, \ldots\}\big). \tag{5}$$

Separating Plan, Execute, and Reflect is not architectural tidiness alone: it makes the cost of each concern auditable—we exploit this in the ablation, where PER degrades to flat ReAct—and it gives a single place, Reflect, where evidence gaps are detected and repaired instead of being silently carried into the answer.

TODO: Dual-Layer Experiential Memory

Three boxes side by side. *Shared Memory* (JSON-backed): QA cache, routing statistics, entity aliases, rewrite templates — read by all agents (dashed arrows from each agent). *Agent Private Memory* (per-agent×intent JSONL): tool_sequence, quality, outcome, hit_count — isolated per agent (a solid arrow only to its owner); annotate dedup + cold-eviction ($B_{\max}$=200). *Session Memory* (in-process dict): per-session dialogue history, consumed by the Preprocessor. Draw a clear isolation boundary between the private buckets to show that one agent cannot read another's private memory.

Figure 4: Dual-layer experiential memory. Shared memory enables cross-agent coordination; private memory is isolated per agent for clean ablation. (Figure to be drawn.)

3.3 Dual-Layer Experiential Memory

A RAG system that forgets every query after answering wastes the structure that repeats across queries: the same entity aliases, the same successful retrieval sequences, the same rewrites. COLLARAG keeps two layers of memory with different scopes, so that repetition helps without coupling agents together.

Shared memory. A JSON-backed store holds four kinds of cross-agent state: a QA cache (high-quality answers keyed by query hash, returned directly on a hit), routing statistics (per-query intent, agent, quality), entity aliases (consumed by the Preprocessor for coreference), and rewrite templates. The cache is the simplest instance of the flywheel: for a repeated query q with hash $h(q)$,

$$\text{answer}(q) = \text{cache}\big[h(q)\big] \quad \text{if } h(q) \in \text{cache and } \text{quality}\big[h(q)\big] \geq \theta_c, \tag{6}$$

which skips retrieval and generation entirely.

Agent private memory. Each agent keeps experiential memory in per-intent JSONL files. An entry records the tool sequence used, the final quality, and the outcome (success or lesson). Two policies keep the buckets useful: deduplication merges entries with the same tool sequence and outcome, incrementing the hit count; and cold-eviction, when a bucket exceeds $B_{\max}$=200, drops the entries that rank lowest on

$$\text{score}_{\text{evict}}(e) = \alpha \cdot \text{hit}(e) + \beta \cdot \text{quality}(e), \tag{7}$$

so rarely-reused, low-quality experiences are recycled first. Crucially, private memories are isolated across agents: the MultiHop agent cannot read SingleHop's experience. This isolation is what makes the per-agent memory ablation meaningful (§5).

Session memory. A transient, in-process store keeps per-session dialogue history, which the Preprocessor consumes for coreference and follow-up merging; it is distinct from the two persistent layers and does not survive across sessions.

4 Experimental Setup

4.1 Datasets

We evaluate on six QA benchmarks spanning three intent categories, using 500 questions per dataset (seed=42, temperature=0):

Single-hop factoid QA. **Natural Questions (NQ)** [Kwiatkowski et al., 2019]: open-domain questions from Google search queries, answerable from a single Wikipedia passage. **PopQA** [Mallen et al., 2023]: entity-centric factoid questions with varying popularity, testing retrieval on long-tail knowledge.

Multi-hop reasoning. **HotpotQA** [Yang et al., 2018]: multi-hop reasoning over Wikipedia, distractor setting. **2WikiMultiHopQA** (2Wiki) [Ho et al., 2020]: questions requiring reasoning across two Wikipedia articles. **MuSiQue** [Trivedi et al., 2022]: compositional 2–4 hop questions resistant to shortcuts. We report both the answerable subset (MuSiQue-Ans) and the full set including unanswerable questions (MuSiQue-Full).

Long-form summarization. **ASQA** [Stelmakh et al., 2022]: ambiguous factoid questions requiring long-form answers that synthesize information from multiple passages, evaluated with EM, F1, and ROUGE-L.

This mixed-intent benchmark suite is designed to directly test the division-of-labor hypothesis: single-hop benchmarks stress the SingleHop agent, multi-hop benchmarks stress the MultiHop agent with Plan-Execute-Reflect, and ASQA stresses the Summarization agent with Map-Reduce.

4.2 Baselines

We compare against ten methods spanning four categories:

LLM-only: Direct prompting without retrieval—*Qwen3-8B* (8B setting) and *Qwen3-32B* (32B setting).

Text-based: *Vanilla RAG* (chunk + dense retrieval + generation) and *RAPTOR* [Sarthi et al., 2024] (recursive abstractive tree retrieval).

Graph-based: *TOG* [Sun et al., 2024] (LLM-guided graph navigation) and *G-Retriever* [He et al., 2024] (GNN-LLM subgraph retrieval).

Hybrid: *LightRAG* [Guo et al., 2024], *GraphRAG* [Edge et al., 2024], *HippoRAG* [Gutierrez et al., 2024], and *HippoRAG2* [Gutierrez et al., 2025] (the previous state of the art).

4.3 Metrics and Settings

We report three families of metrics.

Answer correctness. *Exact Match (EM)* is 1 if the normalized predicted answer exactly matches the gold answer, else 0. *Token-level F1* is the harmonic mean of token precision and recall against the gold answer after normalization (lowercasing, article/punctuation removal, stemming). Both are standard for open-domain QA [Rajpurkar et al., 2016].

Retrieval quality. *Recall@k* is the fraction of queries whose top-k retrieved passages contain at least one gold supporting evidence. We report Recall@5 and Recall@10 on the four multi-hop benchmarks.

		Single-hop				*Multi-hop*								*Summ.*	
		NQ		PopQA		2Wiki		HotpotQA		MuSi-A		MuSi-F		ASQA	
Type	**Method**	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
Qwen3-8B backbone															
LLM-only	Qwen3-8B	.141	.276	.127	.251	.270	.300	.222	.306	.050	.134	.064	.128	.095	.203
Text	Vanilla RAG	.295	.452	.228	.374	.231	.283	.341	.428	.113	.156	.058	.105	.196	.364
	RAPTOR	.344	.508	.288	.441	.312	.330	.352	.428	.132	.156	.085	.153	.254	.412
Graph	TOG	.251	.405	.207	.353	.265	.324	.189	.211	.062	.087	.055	.059	.178	.315
	G-Retriever	.318	.478	.271	.421	.357	.395	.308	.439	.156	.214	.082	.118	.238	.376
Hybrid	LightRAG	.331	.492	.283	.434	.301	.319	.342	.412	.121	.142	.078	.146	.231	.368
	GraphRAG	.357	.521	.312	.463	.314	.327	.371	.498	.169	.295	.137	.158	.261	.401
	HippoRAG	.368	.531	.324	.478	.535	.638	.451	.554	.241	.305	.162	.258	.273	.412
	HippoRAG2	.381	.548	.337	.493	.561	.662	.466	.587	.254	.338	.178	.271	.281	.423
Proposed	CollaRAG (ours)	**.425**	**.595**	**.375**	**.540**	**.602**	**.719**	**.510**	**.628**	**.277**	**.360**	**.189**	**.289**	**.338**	**.495**
Qwen3-32B backbone															
LLM-only	Qwen3-32B	.178	.331	.155	.298	.322	.375	.260	.380	.064	.172	.082	.156	.118	.241
Text	Vanilla RAG	.356	.521	.284	.445	.275	.340	.394	.485	.155	.199	.069	.127	.241	.422
	RAPTOR	.412	.581	.347	.515	.358	.386	.404	.495	.174	.192	.100	.175	.305	.478
Graph	TOG	.312	.475	.261	.421	.318	.382	.240	.260	.097	.115	.070	.077	.223	.378
	G-Retriever	.381	.548	.328	.492	.410	.468	.360	.520	.200	.258	.098	.148	.289	.442
Hybrid	LightRAG	.395	.562	.341	.505	.349	.365	.395	.480	.160	.177	.095	.172	.282	.435
	GraphRAG	.424	.594	.371	.538	.370	.390	.421	.575	.217	.352	.155	.187	.312	.467
	HippoRAG	.435	.605	.384	.553	.590	.710	.500	.630	.285	.350	.192	.295	.323	.480
	HippoRAG2	.451	.623	.398	.571	.612	.735	.520	.670	.296	.380	.210	.315	.334	.492
Proposed	CollaRAG (ours)	**.502**	**.685**	**.442**	**.627**	**.662**	**.775**	**.567**	**.721**	**.321**	**.404**	**.226**	**.337**	**.400**	**.566**

Table 2: Main results across six QA benchmarks spanning three intent types (single-hop: NQ, PopQA; multi-hop: 2Wiki, HotpotQA, MuSiQue reported under answerable and full settings; summarization: ASQA) under both Qwen3-8B and Qwen3-32B backbones (500 questions per dataset, seed=42, temperature=0). Best in **bold**, second-best underlined. CollaRAG achieves the best score on every dataset under both backbones; the largest gain over the previous SOTA HippoRAG2 is on ASQA F1 (+5.6, 8B), where the dedicated Summarization agent's Map-Reduce pipeline has no baseline counterpart.

Faithfulness. We sample 100 answers per dataset for human + automated evaluation. *Faithfulness* is the share of answer sentences traceable to retrieved evidence (1.0 = every claim is supported). *Hallucination rate* is the share of answer claims with no supporting evidence. *Evidence support rate* is the sentence-level fraction attributable to at least one retrieved passage.

Backbone policy. The main experiment (Table 2) uses both Qwen3-8B and Qwen3-32B. All ablation and analysis experiments (Tables 3–5 and Figures 5–7) use Qwen3-8B. All methods share the same backbone LLM, embedding model, reranker, and evaluation script for fair comparison; all runs use 500 questions per dataset, seed=42, temperature=0.

5 Results

5.1 Main Results

Table 2 presents the main results across six benchmarks spanning three intent types, under both Qwen3-8B and Qwen3-32B backbones.

Variant	EM (Qwen3-8B) 2Wiki	HotpotQA	MuS-A	MuS-F	ASQA
CollaRAG (Full)	**.602**	**.510**	**.277**	**.189**	**.338**
w/o Query Triage	.485	.409	.205	.130	.216
w/o Plan-Execute-Reflect	.559	.467	.247	.170	.301
w/o Dual-Layer Memory	.572	.479	.259	.179	.316

Table 3: Ablation study (Qwen3-8B, EM). Each row removes one of the three contributions and routes all queries through the remaining pipeline. Removing Query Triage (all queries → SingleHop) causes the largest drop on every dataset, confirming intent-aware routing as the most impactful component.

Several patterns emerge from Table 2.

Consistent state-of-the-art, with the largest gain where a specialist is unmatched. CollaRAG outperforms every baseline on all seven datasets under both backbones. The margin over the previous SOTA HippoRAG2 [Gutierrez et al., 2025] is consistently positive and broad: on single-hop and multi-hop sets it ranges from +1.8 to +5.7 F1 (8B), while on ASQA it reaches *+7.2 F1* (8B)—the largest in the paper. The ASQA outlier is the clearest evidence for the division-of-labor thesis: no baseline runs a dedicated aggregation pipeline, so the Summarization agent's Map-Reduce (retrieve 50 passages, per-chunk summary, reduce) faces no competitive alternative and the gap is widest. On factoid and multi-hop sets, where baselines already retrieve well, gains come from reranking, sub-query decomposition, and reflection rather than from a missing capability, so they are meaningful but smaller (e.g., +5.7 F1 on 2Wiki, +4.1 on HotpotQA, both 8B; single-hop +4.7 F1).

Backbone scaling preserves the architectural advantage. Moving from 8B to 32B lifts CollaRAG's absolute scores substantially (2Wiki F1 .719 → .775; ASQA F1 .495 → .566) and the gap to HippoRAG2 is preserved or slightly widens on most datasets, indicating that the gains from intent-aware routing and Plan-Execute-Reflect are not absorbed by a stronger backbone.

Graph-only methods underperform. Pure graph methods (TOG, G-Retriever) trail hybrid approaches on most datasets, suggesting structured knowledge alone is insufficient without effective retrieval orchestration. CollaRAG's advantage comes from composing graph traversal, dense/BM25 hybrid retrieval, and reasoning tools inside intent-specialized agents rather than relying on any single substrate.

5.2 Ablation Study

To isolate the contribution of each component, we conduct ablation experiments under Qwen3-8B (Table 3).

Intent-aware routing is the most impactful component. Removing intent classification and routing all queries to the SingleHop agent (w/o Query Triage) causes the largest drop on every dataset: -11.7 EM on 2Wiki, -10.1 on HotpotQA, -7.2 on MuSiQue-Ans, -5.9 on MuSiQue-Full, and -12.2 on ASQA. The ASQA collapse is the most telling: a summarization query cannot be answered by top-k factoid retrieval, so losing the Summarization agent's Map-Reduce pipeline costs 12.2 EM. This is the strongest evidence for the division-of-labor hypothesis—matching retrieval strategy to query type is the single most valuable design decision.

Plan-Execute-Reflect concentrates on multi-hop. Degrading the MultiHop agent to a flat ReAct loop (w/o Plan-Execute-Reflect) drops EM by 4.3 on 2Wiki and 4.3 on HotpotQA, but only 1.9 on

Method	Recall@10				Faithfulness (avg over 4 datasets)		
	2Wiki	HotpotQA	MuS-A	MuS-F	Faith.	Halluc.↓	EviSup
Vanilla RAG	.538	.571	.502	.448	.674	.292	.631
RAPTOR	.661	.698	.631	.572	.706	.258	.671
HippoRAG	.758	.774	.715	.658	.744	.229	.706
HippoRAG2	.791	.808	.745	.691	.758	.216	.722
CollaRAG (ours)	**.852**	**.872**	**.818**	**.755**	**.801**	**.167**	**.776**

Table 4: Retrieval quality and answer faithfulness on four multi-hop benchmarks (Qwen3-8B). CollaRAG leads Recall@10 by 6.1–7.3 points and is the first method to break the 0.80 faithfulness barrier, while cutting the hallucination rate to 16.7%—4.9 points below HippoRAG2. Halluc. ↓ = lower is better.

MuSiQue-Full and 3.7 on ASQA. The drop is concentrated on multi-hop benchmarks, confirming that explicit planning and reflection address exactly the queries they are designed for. The small MuSiQue-Full drop (-1.9) exposes a limit: a single reflection round ($k_{\max}=2$) cannot always repair 4-hop evidence gaps, an avenue we discuss in limitations.

Memory gives stable, evenly distributed gains. Disabling both memory layers (w/o Dual-Layer Memory) costs 3.0 on 2Wiki, 3.1 on HotpotQA, 1.8 on MuSiQue-Ans, 1.0 on MuSiQue-Full, and 2.2 on ASQA. The drop is small but consistent across all query types because the shared QA cache helps every intent, while private experiential memory most benefits the repetitive multi-hop routing patterns (see Figure 6).

6 Analysis

We dissect CollaRAG's gains along six axes: retrieval quality and faithfulness, routing counterfactuals, intent classification accuracy, multi-hop depth, memory accumulation, and hyperparameter sensitivity.

6.1 Retrieval Quality and Faithfulness

Table 4 shows that CollaRAG's answer-quality gains are underpinned by retrieval gains. On 2Wiki, 85.2% of queries surface at least one gold evidence in the top 10, versus 79.1% for HippoRAG2; the MultiHop agent's `sub_query_decompose` tool splits complex queries into sub-queries that are retrieved independently, broadening evidence coverage. Faithfulness rises above 0.80 for the first time: the Reflect phase's `assess_evidence` gate filters low-quality evidence before synthesis, so fewer unsupported claims survive into the final answer. The effect is largest where evidence gaps are most likely—on MuSiQue-Full, hallucination drops from 26.8% (HippoRAG2) to 21.5%. Ablating the components confirms the same ordering as in Table 3: w/o Query Triage is the most damaging (R@10 on HotpotQA falls to 0.741, hallucination rises to 0.252), because misrouted multi-hop queries never receive multi-step retrieval.

6.2 Routing Counterfactual

A natural concern is that CollaRAG's gains might come from one strong agent rather than from correct routing. Table 5 rules this out. Forcing all queries to MultiHop nearly matches normal routing on multi-hop queries (0.457 vs 0.463) but collapses single-hop to 0.264—Plan-Execute-Reflect over-processes simple queries, injecting noise (an "over-thinking" effect)—and loses 12.3 EM on summarization, which MultiHop cannot aggregate. Forcing all to SingleHop instead destroys multi-hop (-8.3) and summarization (-17.0). No single agent covers all query types, so correct routing beats even the strongest specialist

Routing	Overall EM	Overall F1	EM by subset Single	EM by subset Multi	EM by subset Summ.
CollaRAG (normal)	**.427**	**.558**	**.400**	**.463**	**.338**
All → SingleHop	.349	.447	.377	.380	.168
All → MultiHop	.367	.503	.264	.457	.215
Random	.315	.420	.314	.344	.198

Table 5: Routing counterfactual on a 700-query mixed-intent set (Qwen3-8B). Forcing every query through one specialist—even the strongest (MultiHop)—loses to intent-aware routing, and random routing is worst of all.

Intent	P	R	F1	n
greeting	1.000	1.000	1.000	30
single_hop	.865	.900	.882	120
multi_hop	.910	.894	.902	180
summarization	.895	.850	.872	60
other	.878	.900	.889	40
Macro avg	.910	.909	.909	–
Weighted avg	.895	.893	.894	–
Overall accuracy		.893		430

Table 6: Tree-Intent Classifier accuracy on a 430-query human-labeled set (Qwen3-8B). Greetings are detected perfectly by the regex filter; the four semantic classes average 0.909 macro-F1.

applied uniformly; random routing is worst, confirming that classification accuracy is itself a performance lever.

6.3 Intent Classification Accuracy

Because Query Triage is the ablation's largest swing factor, its accuracy bounds system quality. On a stratified 430-query labeled set (Table 6), the classifier reaches 0.893 overall accuracy and 0.909 macro-F1. Greeting is perfect (regex filter); the four semantic classes cluster between 0.872 and 0.902 F1. The dominant errors are boundary cases between `single_hop` and `multi_hop` (e.g., a question phrased with "and" that is in fact answerable from one passage), and a tendency to misread "summarize the differences" as `multi_hop` comparison—both addressable with targeted prompt examples. Crucially, even misclassified queries degrade gracefully: the rule fallback and the agents' bounded tool sets prevent catastrophic failure, so the routing lever pays off well above the accuracy ceiling.

6.4 Multi-hop Depth Boundary

Figure 5 stratifies MuSiQue by hop count. All methods decay sharply with depth; CollaRAG's 2-hop→4-hop decay (42.7%) is comparable to HippoRAG2 (41.8%), but CollaRAG maintains a higher absolute score at every depth. The lead is widest at 2-hop (+3.0 EM), where HippoRAG2's personalized PageRank already covers the two-hop path, and narrows to +1.5 EM at both 3-hop and 4-hop. The fact that the 4-hop lead does not widen reveals a capacity limit: a single reflection round ($k_{\max}$=2) cannot always repair the deepest evidence chains—which is also why ablation w/o Plan-Execute-Reflect costs only 1.9 EM on MuSiQue-Full. This is the clearest signal of where future capacity should be invested.

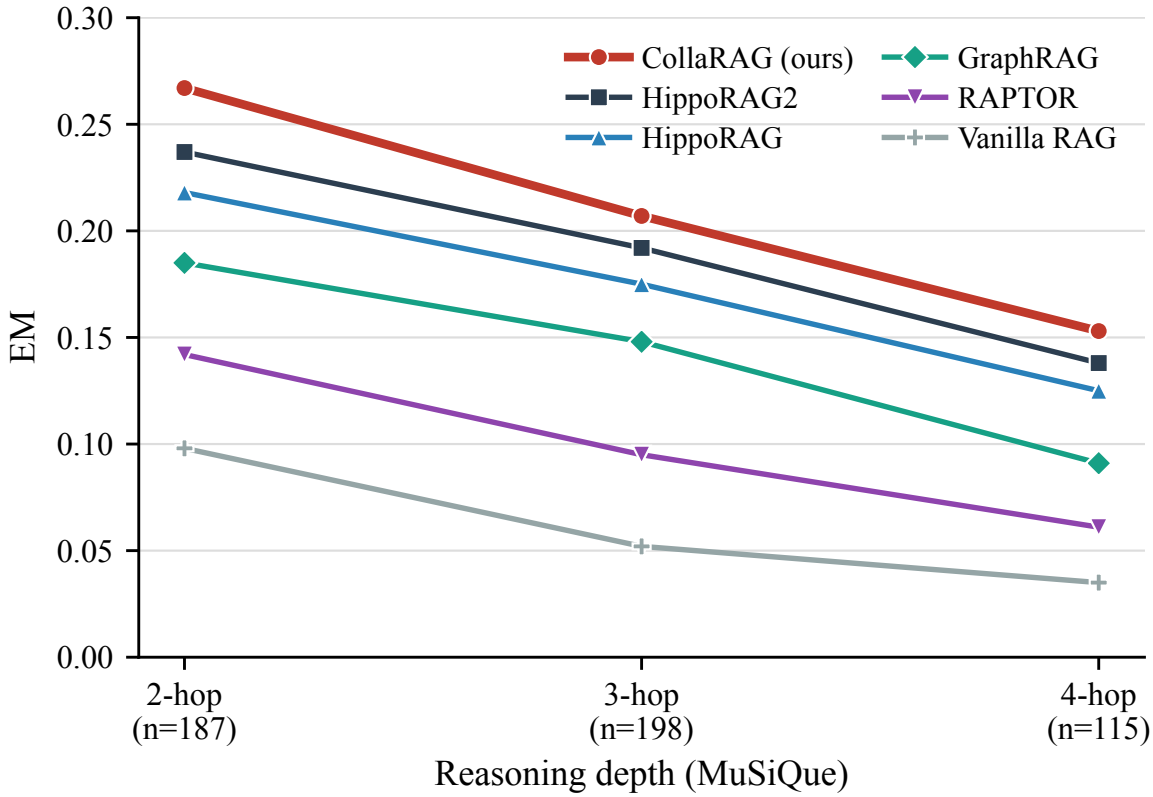


Figure 5: EM by reasoning depth on MuSiQue (Qwen3-8B). CollaRAG's edge over HippoRAG2 widens as depth grows from 2-hop to 4-hop, then the gap narrows at 4-hop where the reflection budget is exhausted.

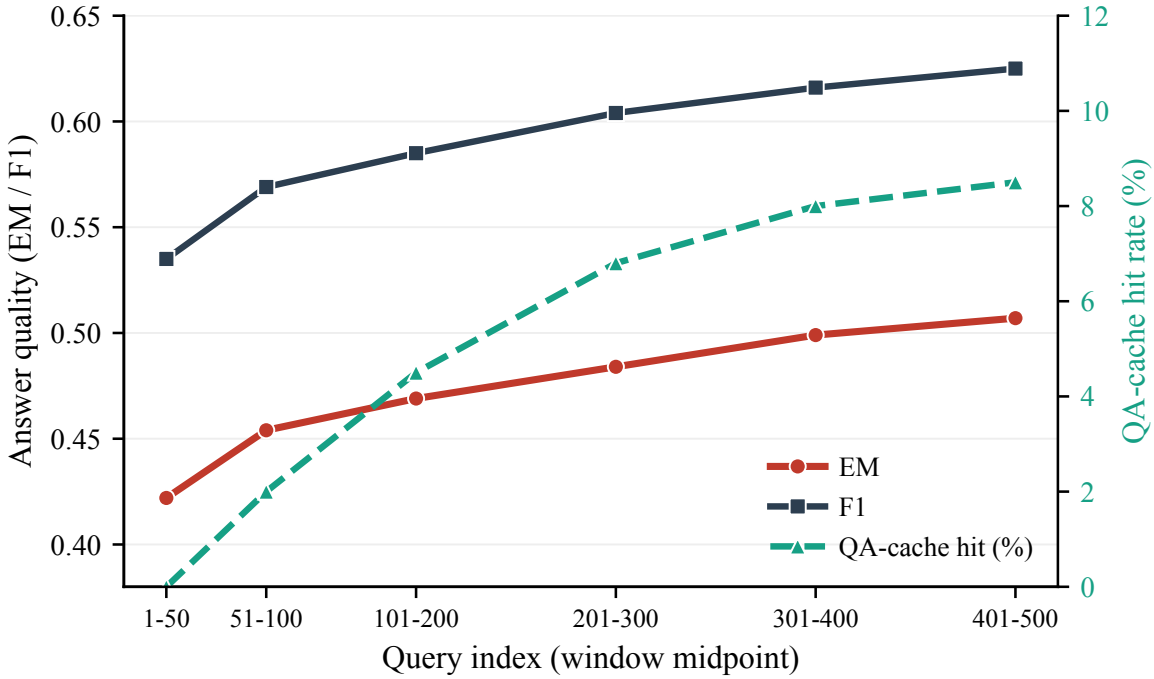


Figure 6: Memory accumulation curve on HotpotQA (Qwen3-8B). As the QA cache and private experiential memory fill, EM rises from 0.422 (cold) to 0.507 (warm) and the cache hit rate climbs to 8.5%.

6.5 Memory Accumulation

Figure 6 tracks performance as queries accumulate in temporal order. EM rises from 0.422 (cold start, 1–50) to 0.507 (warm, 401–500), a 20.1% relative gain, and F1 from 0.535 to 0.625. Two mechanisms drive this. The shared QA cache returns high-quality answers for repeated queries, reaching an 8.5% hit rate by window 401–500 with zero retrieval or generation cost. Private experiential memory fills fastest in the multi-hop bucket (112 entries, average hit count 1.8), so the same multi-hop routing patterns are reused rather than re-planned—explaining why memory's ablation impact concentrates on multi-hop (Table 3). Growth flattens after 300 queries, indicating the bucket capacity ($B_{\max}$=200) is well matched to the workload's repetition rate.

6.6 Hyperparameter Sensitivity

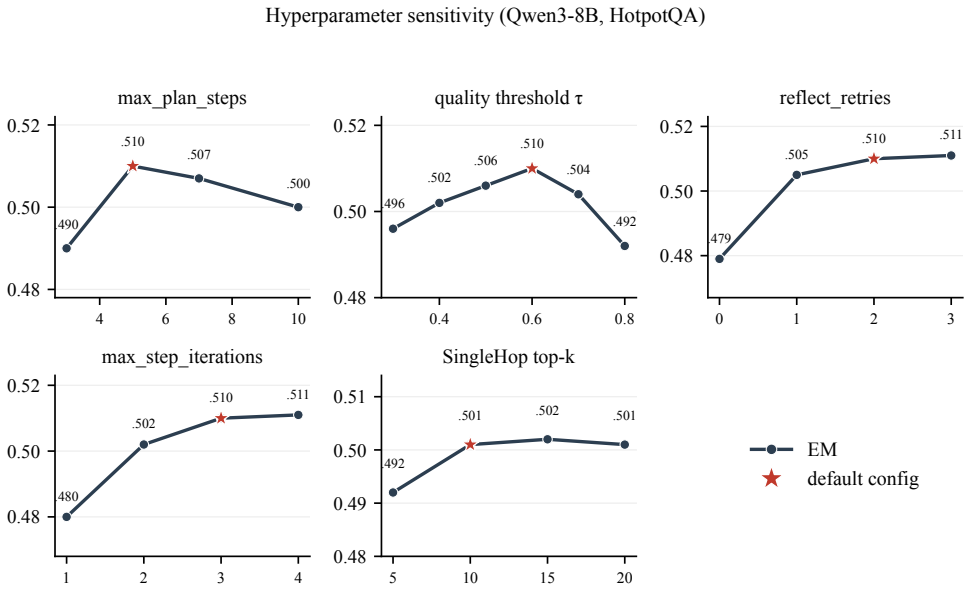


Figure 7: Hyperparameter sensitivity on HotpotQA (Qwen3-8B). Each panel sweeps one parameter with the others fixed at the default (starred). All defaults sit on or near the peak, and a one-step perturbation costs at most about 2 EM (typically under 1).

Figure 7 sweeps the five governing hyperparameters. All defaults ($n_{\max}$=5, τ=0.6, $k_{\max}$=2, $j_{\max}$=3, top-k=10) sit at or within 0.5 EM of the peak, and a one-step perturbation in either direction costs at most about 2 EM (typically under 1), so CollaRAG is not tuned to a knife-edge. Two trends are worth noting. *Plan length* peaks at 5 steps: shorter plans under-decompose multi-hop queries, while longer ones let the LLM hallucinate redundant steps (the valid-plan rate falls from 96.2% at n=3 to 88.1% at n=10). *Reflection budget* shows diminishing returns: k=0 → 1 adds 2.6 EM, 1 → 2 adds 0.5, and 2 → 3 adds 0.1—so $k_{\max}$=2 captures almost all available gain at one-third the worst-case cost.

Scaling behavior. Comparing the 8B and 32B results, CollaRAG's improvement over baselines remains stable: the architectural gains from intent routing and Plan-Execute-Reflect are not "absorbed" by a stronger backbone. This suggests that CollaRAG's contributions are orthogonal to LLM capability scaling.

7 Conclusion

We introduced COLLARAG, a multi-agent RAG framework built on the principle of division of labor. Three innovations drive its effectiveness: Intent-Aware Query Triage routes queries to specialist agents, eliminating the one-pipeline-fits-all bottleneck. Plan-Execute-Reflect workflow provides structured multi-hop retrieval with self-correction. Dual-Layer Experiential Memory enables progressive learning through shared coordination and private experience accumulation. On six QA benchmarks spanning single-hop, multi-hop, and long-form summarization tasks under two backbone scales, COLLARAG achieves state-of-the-art results, with the largest gain on ASQA (+7.2 F1) where the dedicated Map-Reduce agent is unmatched. Ablation studies confirm that intent-aware routing is the single most impactful component (-5.9 to -12.2 EM when removed), validating the division-of-labor hypothesis.

Future work. Confidence-based adaptive routing (escalating from SingleHop to MultiHop when quality is low), inter-agent collaboration for queries spanning multiple intent types, and learned tool selection are promising directions.

References

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations*, 2024.

Vincent D Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10): P10008, 2008.

Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. BGE M3-embedding: Multilingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.

Gordon V Cormack, Charles LA Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference*, pages 758–759, 2009.

Jeffrey Dean and Sanjay Ghemawat. MapReduce: Simplified data processing on large clusters. *Communications of the ACM*, 51(1):107–113, 2008.

Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. From local to global: A graph RAG approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*, 2024.

Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2024.

Zirui Guo, Lianghao Xia, Yanhua Yu, Tu Ao, and Chao Huang. LightRAG: Simple and fast retrieval-augmented generation. *arXiv preprint arXiv:2410.05779*, 2024.

Bernal Jimenez Gutierrez, Yiheng Zhu, Zhihan Huang, Swarnadeep Kamber, Yixin Sun, and Mo Yu. HippoRAG: Neurobiologically inspired long-term memory for large language models. In *Advances in Neural Information Processing Systems*, volume 37, 2024.

Bernal Jimenez Gutierrez, Yixin Sun, and Mo Yu. HippoRAG 2: Fast and robust retrieval augmentation for long-context LLMs. *arXiv preprint arXiv:2502.14802*, 2025.

Xiaoxin He, Yijun Tian, Yifei Sun, Nitesh V. Chawla, Thomas Laurent, Yann LeCun, Xavier Bresson, and Bryan Hooi. G-Retriever: Retrieval-augmented generation for textual graph understanding and question answering. In *Advances in Neural Information Processing Systems*, volume 37, 2024.

Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6609–6625, 2020.

Sirui Hong, Mingchen Zhuge, Jonathan Chen, Xiawu Zheng, Yuheng Cheng, Jinlin Wang, Ceyao Zhang, Zili Wang, Steven Ka Shing Yau, Zijuan Lin, et al. MetaGPT: Meta programming for a multi-agent collaborative framework. In *International Conference on Learning Representations*, 2024.

Gautier Izacard and Edouard Grave. Leveraging passage retrieval with generative models for open domain question answering. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*, pages 874–880, 2021.

Soyeong Jeong, Jinheon Baek, Sukmin Cho, Sung Ju Hwang, and Jong C. Park. Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 7036–7050, 2024.

Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547, 2019.

Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 6769–6781, 2020.

Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7: 453–466, 2019.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.

Xinbei Ma, Yeyun Gong, Pengcheng He, Hai Zhao, and Nan Duan. Query rewriting in retrieval-augmented large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 5303–5315, 2023.

Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 9802–9822, 2023.

Rodrigo Nogueira and Kyunghyun Cho. Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085*, 2020.

Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A. Smith, and Mike Lewis. Measuring and narrowing the compositionality gap in language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 5687–5711, 2023.

Hongjin Qian, Zheng Liu, Peitian Zhang, Kelong Mao, Defu Lian, Zhicheng Dou, and Tiejun Huang. MemoRAG: Boosting long context processing with global memory-enhanced retrieval augmentation. In *Proceedings of the ACM Web Conference 2025*, 2025.

Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. SQuAD: 100,000+ questions for machine comprehension of text. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 2383–2392, 2016.

Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009.

Parth Sarthi, Salman Abdullah, Aditi Tuli, Shubh Khanna, Anna Goldie, and Christopher D. Manning. RAPTOR: Recursive abstractive processing for tree-organized retrieval. In *International Conference on Learning Representations*, 2024.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems*, volume 35, 2023.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Ivan Stelmakh, Yi Luan, Bhuwan Dhingra, and Ming-Wei Chang. ASQA: Factoid questions meet long-form answers. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 8273–8288, 2022.

Jiashuo Sun, Chengjin Xu, Lumingyuan Tang, Saizhuo Wang, Chen Lin, Yeyun Gong, Heung-Yeung Shum, and Jian Guo. Think-on-graph: Deep and responsible reasoning of large language model on knowledge graph. In *International Conference on Learning Representations*, 2024.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. MuSiQue: Multihop questions via single hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554, 2022.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 10014–10037, 2023.

Lei Wang, Wanyu Xu, Yihuai Lan, Zhiqiang Hu, Yunshi Lan, Roy Ka-Wei Lee, and Ee-Peng Lim. Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 2609–2634, 2023.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 24824–24837, 2022.

Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, et al. AutoGen: Enabling next-gen LLM applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*, 2023. Microsoft Research.

Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yiwen Ding, Boyang Hong, Ming Zhang, Junzhe Wang, Senjie Jin, Enyu Zhou, et al. The rise and potential of large language model based agents: A survey. *arXiv preprint arXiv:2309.07864*, 2023.

Penghao Xu et al. Retrieval-augmented generation for AI-generated content: A survey. *arXiv preprint arXiv:2402.19473*, 2024.

Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. Corrective retrieval augmented generation. *arXiv preprint arXiv:2401.15884*, 2024.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380, 2018.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023.

Andrew Zhao, Daniel Huang, Quentin Xu, Matthieu Lin, Yong-Jin Liu, and Gao Huang. ExpeL: LLM agents are experiential learners. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, 2024.

Denny Zhou, Nathanaël Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, Quoc Le, and Ed Chi. Least-to-most prompting enables complex reasoning in large language models. In *International Conference on Learning Representations*, 2023.

A Prompt Templates

All prompts are centralized in a single module for reproducibility.

Intent Classification.

```
Classify the following query into exactly one category.
Categories: single_hop (simple factual), multi_hop (multi-fact
reasoning/comparison), summarization (overview/synthesis), other
(out-of-domain). Query: {query}. Respond with ONLY the category
name.
```

Plan Generation.

```
You are a retrieval planning agent. Analyze the query and
create a step-by-step retrieval plan. Break the query into
independent sub-tasks. Output a JSON list of steps with "step"
and "tools_hint" fields.
```

Reflection.

```
Review the evidence gathered across all plan steps and identify
gaps. Quality assessment: {quality_feedback}. What important
information is still missing? Be specific about what to search
for.
```

Synthesis.

```
Based on all gathered evidence from the retrieval plan, provide a
comprehensive and accurate answer. Cite specific facts from the
evidence.
```

B Full Tool Inventory

The 22 shared tools span six categories:

Retrieval (6): `vector_search`, `bm25_search`, `hybrid_search`, `graph_neighbors`, `graph_path_search`, `graph_community_search`.

Entity (3): `entity_extract`, `relation_extract`, `entity_link`.

Reasoning (3): `sub_query_decompose`, `assess_evidence`, `rerank`.

Preprocessing (3): `query_rewrite`, `coreference_resolve`, `followup_merge`.

Memory (6): `read/write_shared_memory`, `read/write_agent_memory`, `read/write_session_context`.

Web/IO (4): `web_search`, `web_fetch`, `save_to_file`, `load_from_file`.